

Quiz 12

STAT 109: Introductory Biostatistics

Quiz 12

Quiz 12 practice problems

These practice problems cover Lecture 31 (study design and inference language: random sample vs convenience sample; randomized experiment vs observational study; causation, confounding, and chance), Lecture 32 (one-way ANOVA concepts and interpreting overall F-test output), and Lecture 33 (chi-square test of independence concepts and interpreting overall test output).

This quiz is **concepts only**: no hand-calculation is required.

Instructions

- Work without notes first, then check your work.
 - For multiple-choice items, choose the best single answer.
 - Unless a question says otherwise, interpret significance at $\alpha = 0.05$.
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Part A: Independence in study design (paired or not paired)

For each scenario, classify whether observations are **paired/dependent** or **independent/not paired**.

A1.

Thirty students have their resting heart rate measured **before** and **after** a 6-week exercise program. Their measurements before and after are:

- a) Paired/dependent by student
 - b) Independent/not paired within students
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A2.

Two separate sections of STAT 109 take different versions of a quiz. Each student appears in only one section. The quiz scores are:

- a) Paired/dependent between the sections
 - b) Independent/not paired between the sections
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A3.

Researchers match each smoker with one nonsmoker of similar age and sex, then compare lung function between matched pairs. The measurements of lung function are:

- a) Paired/dependent between the groups of smokers and non-smokers

- b) Independent/not paired between the groups of smokers and non-smokers
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A4.

Plant height is measured for seedlings randomly assigned to fertilizer A, B, or C, with each seedling receiving only one fertilizer. The plant height data is:

- a) Paired/dependent between the fertilizer types
- b) Independent/not paired between the fertilizer types
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Part B: Descriptive evidence for ANOVA and chi-square

B1. One-way ANOVA visual evidence

Suppose boxplots are used to compare blood-pressure reduction across three treatments. The medians are visibly different, and one group is shifted noticeably higher than the other two.

Which statement is best?

- a) The boxplot is descriptive evidence of at least one group mean difference.
- b) The boxplot proves all population means are different.
- c) The boxplot shows no evidence of any differences.
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B2. One-way ANOVA overall p-value from R output

You fit `fit <- lm(y ~ group, data = dat)` and run `anova(fit)`. The output shows a p-value = 0.012.

At `alpha = 0.05`, which interpretation is best?

- a) Reject H_0 ; the data provide evidence that at least one population mean differs beyond chance variation under the model assumptions.
- b) Fail to reject H_0 ; all means are exactly equal.
- c) Reject H_0 ; therefore every pair of means is significantly different.
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B3. Chi-square descriptive evidence

A 100% stacked bar chart shows disease status (**yes/no**) by exposure group (**low/medium/high**). The segment proportions differ clearly across exposure groups.

Which statement is best?

- a) This is descriptive evidence variables may not be independent.
- b) This proves causation by exposure.
- c) This shows the two variables must be independent.

B4. Chi-square overall p-value from R output

You run `chisq.test(tab)` and see `p-value = 0.18`.

At `alpha = 0.05`, which conclusion is best?

- a) Reject H_0 ; there is strong evidence of association.
 - b) Fail to reject H_0 ; sample data are compatible with independence (not enough evidence of association).
 - c) Accept H_0 as true with certainty.
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Part C: Interpreting R code output (conceptual)

C1. ANOVA null and alternative

For a one-way ANOVA with 4 groups, which is the correct null?

- a) H_0 : all four population means are equal.
 - b) H_0 : all four sample means are equal.
 - c) H_0 : at least one population mean differs.
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C2. Chi-square test null and alternative

For a chi-square test of independence between treatment arm and side-effect category:

- a) H_0 : treatment arm and side-effect category are independent in the population.
 - b) H_0 : treatment arm causes side effects.
 - c) H_0 : every cell count in the sample table is equal.
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C3. Interpreting overall p-value language

Complete the statement:

“A small p-value means that, **if the null were true**, data this extreme would be _____.”

- a) likely
 - b) uncommon (unlikely)
 - c) impossible
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C4. “By chance” language

Which sentence is best statistical wording when $p = 0.003$ for an ANOVA overall test?

- a) “There is strong evidence against equal means; such differences would be unlikely by chance alone if all means were

equal.”

- b) “The null is impossible and chance never plays a role.”
 - c) “This proves treatment mechanism with no assumptions.”
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Part D: Generalizing to a population (sampling design)

D1.

A researcher recruits volunteers from people walking into one gym between 5-6 pm and surveys protein intake. Can the results be safely generalized to all adults in the city?

- a) Yes, because sample size is all that matters.
 - b) Not reliably; this is a convenience sample and may not represent the city population.
 - c) Yes, because any sample supports broad generalization.
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D2.

A simple random sample (SRS) is drawn from a complete city resident list. Which statement is best?

- a) Generalization to that city population is as safe as we can manage from limited information
 - b) No generalization is ever possible, even from SRS.
 - c) SRS automatically proves causation.
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Part E: Causation, confounding, and chance (study type)

E1.

In an **RCT**, participants are randomly assigned to drug vs placebo. The drug group has a lower mean symptom score and a hypothesis test comparing the population means has a very small p-value.

Which interpretation is best?

- a) Random assignment helps support a causal interpretation
 - b) Because chance exists, random assignment gives no advantage.
 - c) RCTs cannot address causation.
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E2.

In an **observational** study, soda intake is positively associated with BMI.

Which interpretation is best?

- a) The association could reflect causation, confounding or chance; causation is not established by this design alone.
 - b) Observational association proves soda causes higher BMI.
 - c) Confounding is impossible in observational studies.
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E3.

A convenience sample observational study reports a significant association ($p < 0.01$) between supplement use and fewer colds.

Best interpretation:

- a) Significant p-value alone proves causal effect for everyone.
- b) Evidence of association in the sampled data, but external validity and confounding remain concerns.
- c) Because p is small, sampling design no longer matters.